HITL (Human-in-the-Loop)

What is HITL?

HITL = Human-in-the-Loop

AI makes a trading decision → **Human reviews it** → Execute after approval

High-risk decisions need human oversight — we can't delegate everything to AI.

Why HITL?

Current pipeline: **User input → Multi-Agent Analysis → PM Decision → Execute directly**

Risks:

- PM suddenly suggests 60% position in a single stock
- PM recommends a trade with only 40% confidence

Gap: No approval mechanism — AI output is the final decision.

Architecture

```
PM outputs decision
     │
     ▼
┌──────────────┐
│  Rule Engine │  ← Evaluates if human review is needed
└──────────────┘
       │
       │
  ┌────┴────┐
  ▼         ▼
Auto-pass   Approval Queue → Human Review → Approve / Reject / Modify
```

PM decision → Rule check → Create approval if needed → Human review → Execute or skip

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Three Trigger Rules

Rule	Threshold	Rationale
Position Change	> 20%	Prevent sudden large portfolio shifts
Low Confidence	< 0.5	Human should intervene when AI is unsure
High Concentration	> 30%	Avoid over-concentration in a single ticker

OR logic: any single rule triggers approval

HOLD + 0% position → no-op, auto-passed

State Machine

```
         PENDING
            │
   ┌────┬───┼────┐
   ▼    ▼   ▼    ▼
APPROVED REJECTED MODIFIED TIMED_OUT
   │       │       │        │
   ▼       ▼       ▼        ▼
Execute  Skip   Execute    Skip
(original)     (modified)
```

One-way transitions only — no rollback, invalid transitions rejected

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Backend Implementation

New `hitl/` module (5 files)

File	Purpose
`rules.py`	Rule engine: evaluates if approval is needed
`state_machine.py`	State machine: manages status transitions
`executor.py`	Executor: outputs final execution parameters
`models.py`	Pydantic data models
`server/routes/approvals.py`	5 REST API endpoints

Integration: HITL check inserted after PM output in `analyze.py`, creates approval record + returns status via SSE

Database Design

Table: `approvals` (stored in `{strategy_id}.db`)

Field	Description
`ticker`	Stock symbol
`original_action` / `original_target_position_pct`	Original decision
`original_confidence`	Confidence score
`triggered_rules_json`	Triggered rules

Field	Description
`status`	pending / approved / rejected / modified
`reviewer` / `reviewer_notes`	Reviewer info
`modified_action` / `modified_target_position_pct`	Modified parameters

Frontend

`frontend/app/approvals/page.tsx`

- **Pending Tab**: Approval cards (ticker, action, position, confidence, rules, PM report)
- **History Tab**: Processed approvals
- **Actions**: Approve / Reject / Modify (inline form)
- **Auto-refresh**: Polling every 10 seconds
- **PM Report**: ReactMarkdown rendering

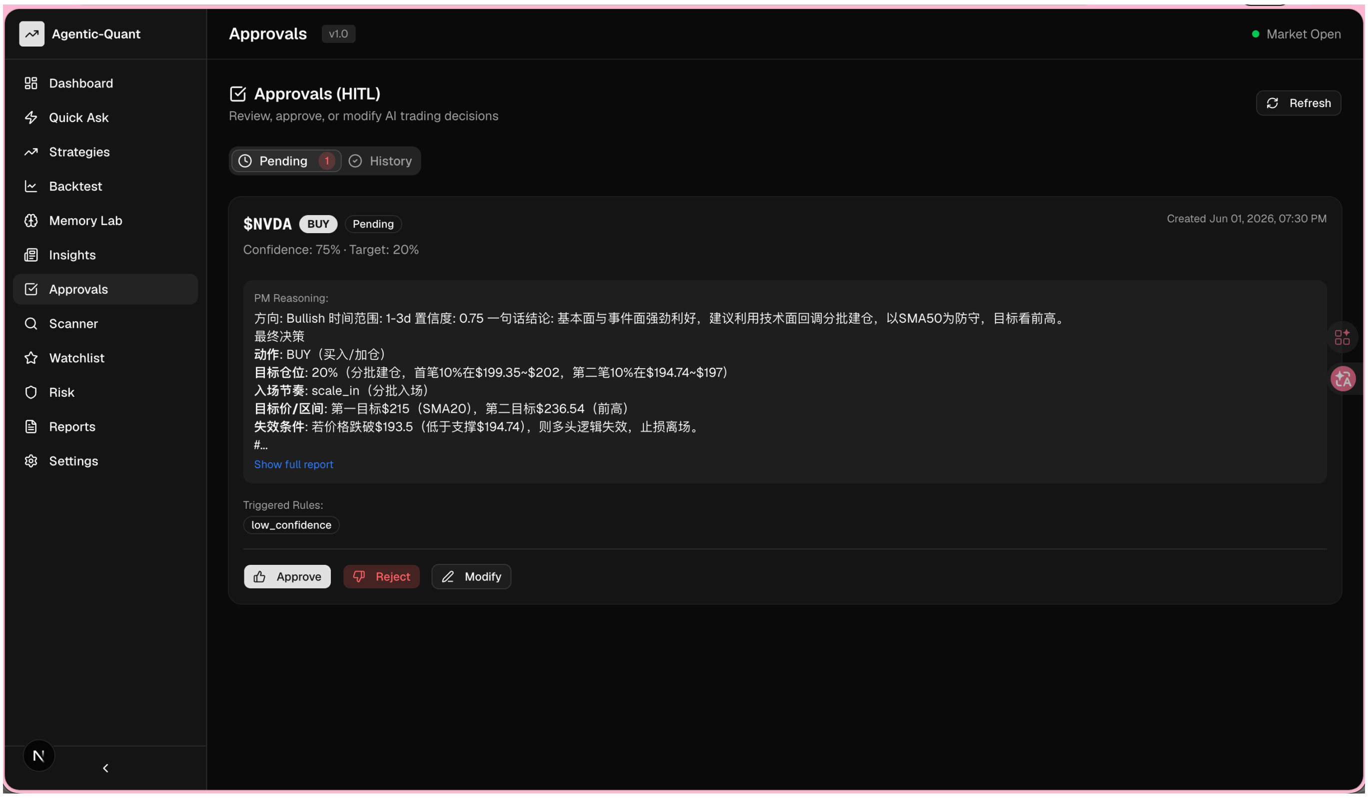
Demo

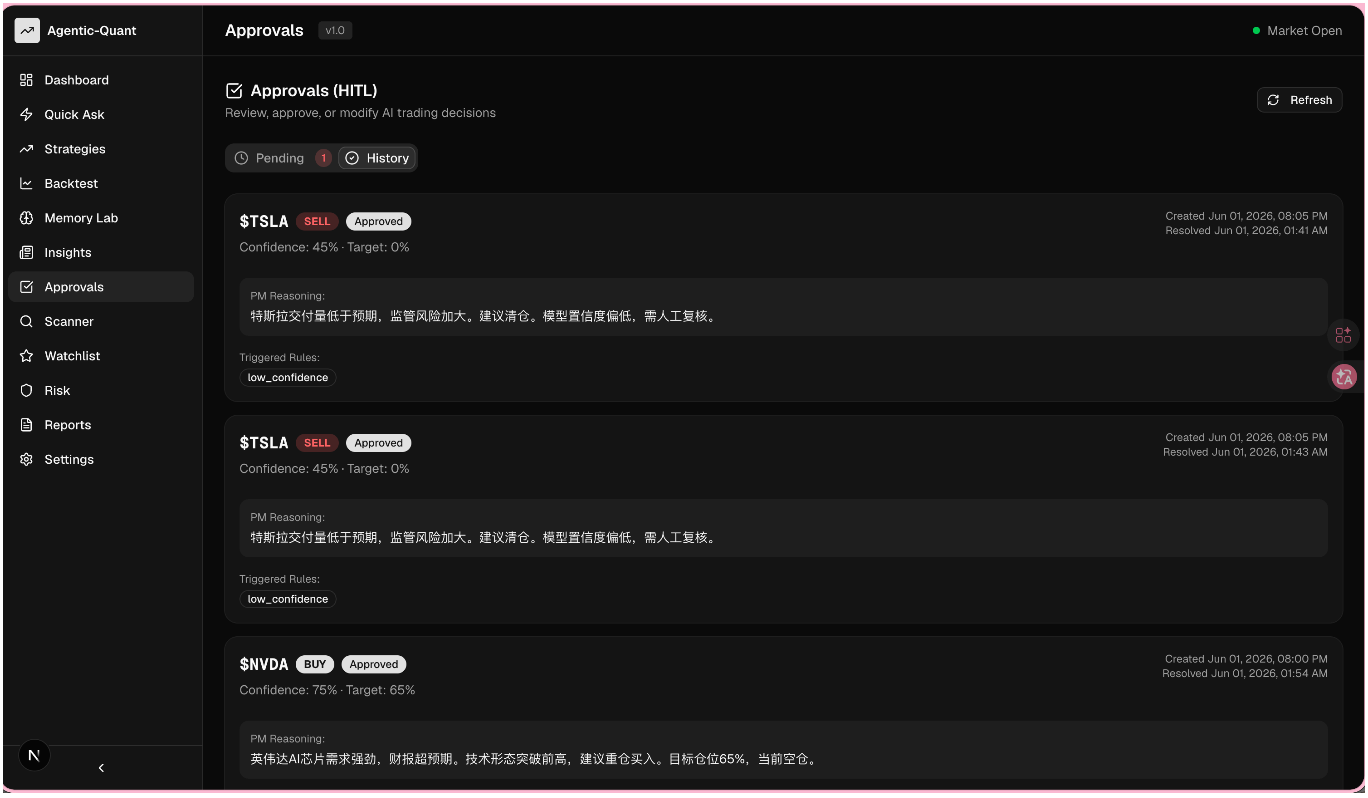
Scenario: Analyze NVDA → PM suggests BUY 65%

Triggered rules: position_change + high_concentration

Card shows: $NVDA · BUY · 65% · Confidence 75% · PM full report

Actions: Approve (execute) / Reject (skip) / Modify (adjust then execute)





API Endpoints

Method	Path	Purpose
GET	`/api/approvals?status=pending`	List pending approvals
GET	`/api/approvals/{id}`	Get single approval
POST	`/api/approvals/{id}/approve`	Approve
POST	`/api/approvals/{id}/reject`	Reject
POST	`/api/approvals/{id}/modify`	Approve with modification

All endpoints tested and verified.

Limitations & Future Work

Limitations:

- Thresholds are hardcoded, not wired to strategy config
- Approval timeout field reserved but not implemented
- Executor only logs results, doesn't call Broker (Gap C)

Future Work:

- Integrate with StrategyConfig for per-strategy HITL settings
- Implement auto-timeout handling + notifications
- Build detailed approval detail page

Summary

- Complete HITL module: Rule Engine + State Machine + Executor
- 5 REST API endpoints + Frontend approval page
- Integrated into the analysis pipeline

~1,150 lines of code, 7 new files, 6 modified files

Every high-risk AI decision now requires human review before execution.

Q&A

Thank you. Questions are welcome.